

Deciphering Braille: EEG based Deep learning for character recognition

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Abstract—The study of experience-based plasticity in the human cortex, especially in those who lost eyesight at birth or early in life, provides valuable insights into the brain's ability to adjust to sensory loss. This study examines how the brain reorganises to map tactile information, such as Braille, by shifting functions from vision to touch processing areas. The work employs EEG to track the transition from sensory to perceptual representations, revealing spatiotemporal changes. These findings have major implications for Brain-Computer Interface (BCI) technologies. Understanding how the brain repurposes regions for different sensory modalities might help researchers develop BCI designs to compensate sensory deficiencies, improve neural signal interpretation, and guide user behaviour effectively. Integrating neural adaptation insights with BCI development can enhance assistive technology to better meet the needs of individuals with sensory disorders. This study explores how the brain responds to braille letter recognition, a crucial aspect of tactile reading for persons with visual impairments. We used statistical and deep learning based techniques to categorise braille representations in the brain. LSTM achieved 88.9% accuracy in categorising left-vs-right hands, while CNN scored 20.3% accuracy across eight classes. Deep learning outperformed machine learning method across both the scenarios.

Index Terms—Brain Computer Interface, Neural plasticity, EEG decoding, Braille character recognition,

I. INTRODUCTION

Braille is a vital tool used by people who are blind or have low vision for literacy, independence, and inclusivity. This writing system is tactile. However, there exist further obstacles to effectively learning and mastering Braille, like limited access to resources, inadequate training for teachers, etc. Studies conducted in the field of neuroscience are yielding important insights into how the brain interprets touch cues and adjusts to vision loss. This information can help visually impaired people make the most of their abilities and enhance projects promoting Braille reading. Studies have indicated that learning and adjusting to new sensory modalities, like reading Braille, are significantly influenced by neuroplasticity, the brain's ability to reorganize itself[1][3].

BCR systems use image processing, machine learning, and algorithms to interpret Braille text, converting it into digital formats or synthesized speech. When combined with brain-

computer interface (BCI) technology, BCR can enhance interaction in a more natural and efficient manner. The collaboration between BCR and BCI has the potential to transform assistive technologies and enhance the independence of individuals with visual impairments in the community. Recent neuroscience research has demonstrated that the brain can learn new skills and adapt even in the absence of visual information.

Neuroscientific studies have shown that blind people exhibit remarkable neuroplasticity, as seen by the activation of the visual cortex and tactile processing regions during Braille reading. Cross-modal activation improves reading comprehension by making tactile stimuli easier to understand. Understanding these neural mechanisms can inform customized Braille training programs that leverage each learner's unique neuroplastic capabilities. Insights from neuroscience can also guide the development of enhanced instructional techniques, assessment tools, and technology-assisted Braille devices that align with how the brain processes tactile information. By bridging the fields of neuroscience and Braille education, we can create more effective, evidence-based strategies to support literacy, independence, and quality of life for the visually impaired community[3].

Our study was to find the answers to certain hypothesis:

1. Are neural oscillatory patterns differentially represented for left-hand and right-hand Braille reading tasks in visually impaired individuals?
2. Are braille characters uniquely encoded in the brain ?
3. Can deep learning based models capture the latent representations of braille characters from EEG signals compared to traditional machine learning models?

To address these research questions we made use of an existing OpenNEURO Braille letters dataset[4]. In this study, we explored a machine learning model, Random Forest and two deep learning models, Convolutional Neural Networks (CNNs) with ResNet architecture, and Long Short-Term Memory (LSTM) networks to decode multiple brain network patterns.

II. RELATED WORK

The exploration of tactile perception and braille reading has been the subject of numerous studies in cognitive neuroscience, particularly focusing on the neural mechanisms and

processing pathways involved. The following review of related works provides context and background for the present study.

A. Tactile Perception and Braille Reading

In blind individuals reading Braille, several specific brain regions are activated, demonstrating the brain's remarkable adaptability. Key areas include the Visual Word Form Area (VWFA), located in the left occipitotemporal cortex, which is crucial for reading and is similarly activated in both blind and sighted readers, suggesting its role in processing written language regardless of sensory modality [6]. The lateral occipital cortex (LOC), which is a component of the visual cortex responsible for higher-level processing, exhibits significant activation during Braille reading [5]. This implies a role for the LOC in integrating tactile and visual information. Furthermore, the primary visual cortex's activation indicates that the brain may modify its visual processing regions to accommodate tactile reading tasks [8]. The inferior frontal cortex (IFC), connected to higher order cognitive functions including language processing, and the somatosensory cortex, which analyzes touch information, are two other important regions. The observation of functional connectivity between these regions highlights the complex neural networks responsible for Braille reading and demonstrates the brain's capacity to reorganize in response to sensory deprivation [5][8].

The initial study by Sadato et al. [11] found that blind individuals exhibit activation in the visual cortex when performing tactile tasks, indicating the presence of cross-modal plasticity. This pioneering research [6] has led to further exploration of how the brain adjusts to sensory deprivation and amplifies other senses, such as touch. Building on these findings, Van Boven et al.'s study demonstrated that tasks involving tactile discrimination activate the primary somatosensory cortex (S1) in both sighted and blind individuals, emphasizing the significance of S1 in intricate tactile processing.

A significant study examined the event-related potentials (ERPs) of individuals as they engaged in Braille reading. The researchers focused specifically on the N400 effect, which is associated with semantic processing. The study revealed distinct patterns of event-related potentials (ERPs) in proficient Braille readers as they encountered regular or inconsistent sentence endings. This indicates that the brain processes Braille text in sighted individuals similarly to printed text. Additionally, the research underscored the significance of kinematic data by demonstrating substantial variations in finger motion frequency and velocity in response to sentence congruence. This provided a deeper insight into the mental and physical processes involved in Braille reading.

Furthermore, research utilizing EEG has specifically examined how the brain handles tactile Braille stimuli associated with somatosensory perception. Early findings indicate that reading Braille activates specific brain areas linked to touch perception, suggesting that intricate neural pathways are necessary for the brain to adapt to reading Braille. These findings align with extensive studies on brain plasticity, indicating that

blind individuals' brains can restructure themselves to improve their sense of touch. This offers additional evidence supporting the idea that reading Braille activates the same brain pathways in visually impaired individuals as reading through sight [2].

Overall, the evidence from EEG studies underscores the significance of understanding the neurodynamics of Braille reading. By tracing how Braille information is processed in the brain, researchers can develop more effective training methods and assistive technologies, ultimately improving literacy outcomes for visually impaired individuals. The integration of EEG findings with other neuroimaging techniques, such as fMRI, holds promise for further elucidating the cognitive processes involved in Braille reading and the potential for enhancing brain function through targeted interventions [7][9].

B. Braille and Neural Adaptations

Braille reading involves complex neural adaptations, as shown in studies by Bichel et al. and Amedi et al. These studies used functional magnetic resonance imaging (fMRI) to reveal that proficient braille readers exhibit enhanced activation in somatosensory regions and the occipital cortex [3][4]. However, EEG studies on braille reading are relatively scarce. Our experiment aims to fill this gap by providing fine-grained temporal data on the neural responses to braille stimuli.

Long term training in braille reading has shown to induce structural changes in the brain particularly in the visual cortex which is re purpose for tactile processing. Blind individuals often experience cognitive deficits that can manifest in various ways, particularly in memory and overall cognitive functioning. Research indicates that visual impairment is significantly associated with an increased risk of cognitive decline, particularly among older adults. For instance, studies have shown that individuals with visual impairment exhibit higher rates of cognitive impairment and memory loss compared to their sighted peers, with some studies suggesting that blindness may accelerate cognitive decline (as evidenced by hazard ratios indicating a greater risk of cognitive deterioration in blind individuals) [9][10]. The brain doesn't process many of the information and a major share of the information will be lost in people who are visually impaired (optometrists.org). This will lead to neuro degeneration and cognitive training is one of the most important modality which can be used to retain plasticity effects and prevent the neuro degeneration. It is very important to understand the neural oscillatory mechanisms and how really brain braille reading brings in neuroplasticity changes. Thus we can have efficient and robust BCI in braille reading.

C. Behavioral Correlates of Braille Reading

Studies on behavior have contributed much to our understanding of the cognitive and perceptual aspects of reading Braille, particularly those conducted by researchers such as Millar and Legge et al. [14]. Millar's research emphasizes the importance of hand dominance and reading tactics for learning

to read Braille. For instance, Millar found that right-handed individuals may develop different tactile reading strategies than left-handed individuals, which may have an impact on their overall reading efficiency and degree of comfort with Braille. This suggests that hand dominance influences how successfully people engage with Braille, which in turn influences learning processes and reading outcomes.

The study by Legge et al. also examines several factors that influence the accuracy and speed of Braille reading, highlighting the significance of tactile acuity and letter similarity. Their results indicate a relationship between improved reading accuracy and speed and increased tactile acuity, which is the ability to detect fine details by touch. On the other side, a higher degree of similarity between Braille letters may lead to confusion and slow down reading times since readers could find it difficult to discern between identical characters. This relationship between tactile sensitivity and letter recognition demonstrates how challenging reading Braille is as a cognitive task and how perceptual abilities directly affect literacy in individuals who are visually impaired.

D. Experimental Paradigms in EEG Studies

Studies using electroencephalography (EEG) have greatly advanced our knowledge of the tactile reading mechanisms of Braille. Following the successful models for visual event-related potentials (ERPs) developed by Duncan et al. [22, 26], further research has applied comparable methods to examine Braille reading. Moreover, the amalgamation of EEG and kinematic data has furnished valuable insights into the motor aspects of Braille reading. The findings show that there are significant variations in the frequency and pace of finger movements in relation to sentence congruence. Management of inter-stimulus intervals (ISIs) in Braille research has been further informed by Papanicolaou et al.'s work on auditory ERPs to reduce movement artifacts. Overall, this research enhances our understanding of sensory processing in tactile reading and has broader implications for neuroplasticity and cross-modal sensory integration[15], ultimately paving the way for improved educational strategies and assistive technologies for the visually impaired community.

III. METHODOLOGY

We have taken the Braille letters OpenNEURO EEG dataset for this study. We applied basic preprocessing techniques (FIR filtering) to filter the frequency within meaningful ranges of EEG. We employed a machine learning approach (Random Forest) and two deep learning approaches (Convolutional Neural Network and Long Short-Term Memory network) to decode the brain's electrical activity during Braille reading. The basic flow diagram is shown in Fig 1.

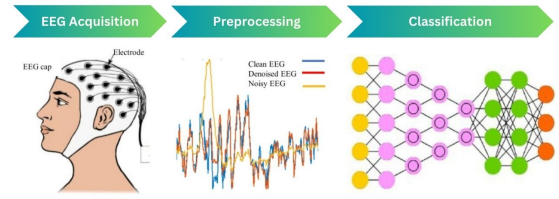


Fig. 1. Basic Flow diagram of the study

A. EEG Dataset & Experiment

The EEG experiment consisted of two sessions involving a total of 11 participants, with 8 of them completing both sessions. Each session lasted 59 minutes, not including breaks. Braille letters were displayed for 500 ms, followed by a 500 ms inter-stimulus interval (ISI) during regular trials, and for 500 ms with an 1100 ms ISI during catch trials to minimize movement contamination. Each of the 16 experimental conditions was presented 170 times per session, interspersed with catch trials, resulting in 2720 regular trials and 541 catch trials per session, for a total of 3261 trials. Due to technical issues, two participants had additional trials, leading to final counts of 190 and 180 repetitions per stimulus. Before the experiment, participants completed a screening task in which they identified Braille letters presented to each hand.

B. EEG data Pre-processing

The dataset used in this study has undergone comprehensive preprocessing to ensure high-quality input for analysis. The data has been processed using a low-pass filter set at 50 Hz to eliminate high-frequency noise. With epochs divided into 100 ms and 999 ms windows, 1100 data points are produced for each epoch. To minimize baseline drift and equalize the data, baseline correction has been implemented. Re-referencing has been done to improve spatial resolution, and artifacts have been found and eliminated using Independent Component Analysis (ICA) with SASICA. Furthermore, to further minimize noise across several channels, multivariate noise normalization has been used. The dataset is guaranteed to be adequately prepared for precise and trustworthy EEG signal analysis thanks to these preprocessing procedures.

C. Classification based on machine learning (Random forest)

EEG signals can be efficiently classified for Braille representation using a Random Forest classifier by utilizing important feature extraction and classification methods. An ensemble of decision trees called the Random Forest, as shown in Fig 2 divides the data recursively according to the best feature and threshold for maximizing information gain, which is usually quantified in terms of entropy or Gini impurity.

$$\text{Gini}(t) = 1 - \sum_{i=1}^c p_i^2 \quad (1)$$

$$\text{Entropy}(S) = - \sum_{i=1}^c p_i \log_2(p_i) \quad (2)$$

The classifier's final prediction is determined by majority voting among the trees.

$$\hat{y} = \text{mode}(\{h_1(X), h_2(X), \dots, h_T(X)\}) \quad (3)$$

where $h_i(X)$ is the prediction made by the i -th tree

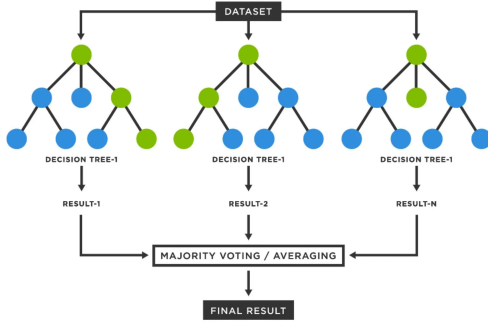


Fig. 2. Architecture of Random Forest

Model performance is evaluated using F1-score, providing a comprehensive measure of the classifier's effectiveness. Our Random Forest classifier was configured with 100 decision trees with the Gini criterion. The maximum depth was set as 'None' allowing the tree to expand until there is a single leaf node. The other parameters were used as the default in the scikit learn library.

D. Classification based on Deep learning

Since the problem we are trying to solve involves nonlinear functions and mappings, the conventional random forest classifier may not be able to approximate those nonlinear complex functions. Hence we hypothesise that a deep learning based approach will be a better model to capture the intrinsic complexities.

1) *CNN*: To classify EEG signals for Braille representation using a Residual Network (ResNet), we first preprocess the EEG data by applying filters to remove noise and segmenting the signals into appropriate time windows. ResNet, a sophisticated variant of Convolutional Neural Networks (CNNs), overcomes the gradient descent problem, particularly the vanishing gradient issue, through residual learning and skip connections. Instead of directly learning the desired mapping, ResNet learns the residuals (the differences) between the input and output, allowing for effective training of very deep networks. The skip connections enable gradients to flow more easily during backpropagation, mitigating the risk of vanishing gradients and facilitating the training of networks with hundreds of layers, while preserving essential information

as it passes through the layers. This architecture leads to improved performance and efficiency in training deep neural networks.

The core component of ResNet is the residual block, which allows the network to learn identity mappings by adding short-cut connections that bypass one or more layers. The operation within a residual block can be represented as

$$y = \mathcal{F}(x, \{\mathbf{W}_i\}) + x, \text{ where } x \text{ is the input, } \mathcal{F}(x, \{\mathbf{W}_i\}) \quad (4)$$

The ResNet architecture begins with an initial convolutional layer, followed by several residual blocks. Each block comprises convolutional layers that include batch normalization and ReLU activation functions, as illustrated in Fig 3.

$$\text{ReLU}(x) = \max(0, x) \quad (5)$$

The output layer uses a softmax activation function for classification,

$$\text{Softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}} \quad (6)$$

The ResNet model is trained using backpropagation to minimize the categorical cross-entropy loss function,

$$\text{CrossEntropy}(y, \hat{y}) = - \sum_i y_i \log(\hat{y}_i) \quad (7)$$

with optimization algorithms such as Adam or SGD.

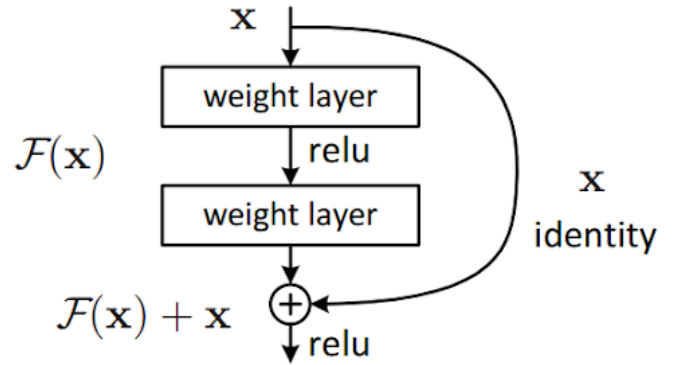


Fig. 3. Architecture of Resnet

We used the ResNet 18 architecture for classification which belongs to the ResNet family, and mitigates the vanishing gradient problem in deep neural networks. The core innovation of ResNet 18 lies in its residual blocks, which incorporate identity shortcut connections bypassing one or more layers.

These shortcuts facilitate the gradient flow during backpropagation, thus enabling the training of substantially deeper networks without performance degradation. The ResNet used BinaryCrossEntropy with Logic Loss and an adams optimizer was used with an initial learning rate of 0.001 and a weight decay of e^{-1} . The learning rate of the model was increased in case the loss has plateaued in order to prevent the gradient being stuck in a local minima.

2) *LSTM*: LSTM, a variant of Recurrent Neural Networks (RNNs), is particularly effective for handling sequence data such as EEG signals because of its capacity to capture long-term dependencies. Long Short-Term Memory (LSTM) networks are specifically designed to manage both short-term and long-term temporal dependencies in sequential data through a sophisticated gating mechanism. This mechanism consists of three types of gates: the input gate, the forget gate, and the output gate, which control the flow of information into and out of the cell state, as depicted in Fig 4. The cell state serves as a memory that retains information over extended periods. By efficiently determining which information to keep or discard at each time step, LSTMs address the vanishing and exploding gradient problems that often affect traditional RNNs.

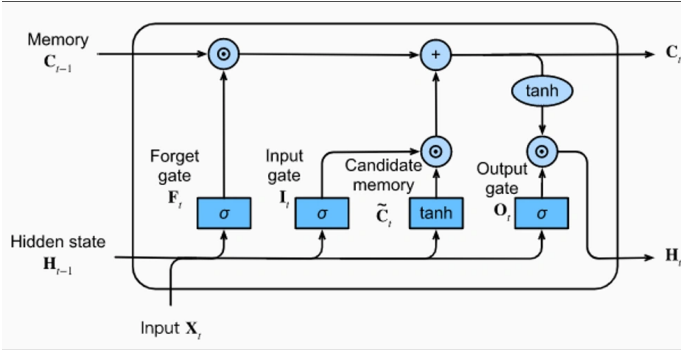


Fig. 4. Architecture of LSTM

LSTMs have three gates:

- **Forget Gate:** This gate identifies the information to be discarded from the cell state. It employs the Sigmoid activation function to assess the extent of memory that should be retained.
- **Input Gate:** This gate determines which new information should be stored in the cell state.

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (8)$$

- **Cell State:** Updates the cell state with new candidate values and information from the forget gate.

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (9)$$

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t \quad (10)$$

- **Output Gate:** Determines which information to pass to the hidden state.

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (11)$$

- **Hidden State:** The final output of the LSTM cell, which also influences the output of the next time step.

$$h_t = o_t \cdot \tanh(C_t) \quad (12)$$

Our LSTM model had an initial learning rate of 0.01 with hidden size of 64 and a dropout of 0.3.

IV. RESULTS, DISCUSSIONS & METHOD COMPARISON

The left vs right hand mapping was observed for Braille reading in people with visual loss. All three classifiers accurately mapped Braille hand from EEG data. However, brain representations overlapped for diverse symbols, and classifiers performed at chance levels when distinguishing separate Braille letters. This is important because it demonstrates the commonality of Braille letter representations in the brain. Additionally, this demonstrates the improvements that can be made to the Braille scripts based on evidence from neural data.

In comparing classification tasks for EEG-based Braille character recognition, the CNN model consistently outperformed both Random Forest and LSTM models across all tasks as shown in Table 1. For binary classification of left vs. right hand, the CNN achieved the highest F1 score at 79%, slightly surpassing the LSTM at 77%, while the Random Forest lagged at 74%.

TABLE I
COMPARISON OF CLASSIFICATION ACCURACIES

Classification Task	Random Forest	CNN	LSTM
left vs right hand	83.3 %	88%	88.9%
8 classes from left hand	13.1%	20.3%	13.47%
8 classes from right hand	13.3%	20.1%	13.9%
8 classes irrespective of hands	13.1%	20.1%	13.5%

In classifying 8 classes from the left hand, the CNN excelled with 20.3% accuracy, whereas the Random Forest and LSTM showed similar performance at 13.1% and 13.47%, respectively. For 8 classes from the right hand, the CNN again led with 20.1%, with the LSTM and Random Forest at 13.9% and 13.3%. When classifying 8 classes irrespective of hands, the CNN maintained the highest accuracy at 20.1%, while the Random Forest and LSTM performed similarly at 13.1% and 13.5%. These results indicate that while the CNN is notably more effective for detailed classifications, the LSTM shows slightly better performance than the Random Forest in some tasks. Overall, CNNs demonstrate superior capability in EEG-based Braille character recognition, highlighting their potential to enhance the accuracy and reliability of assistive technologies for the visually impaired, though improvements are needed for more granular classifications.

The comparisons between sensitivity-specificity for three classifiers are shown in the Fig 5. CNN outperform LSTM and Random Forests in these tasks due to their ability to efficiently capture spatial hierarchies and features through convolution, parameter sharing, and translation invariance, making them better suited for high-dimensional data.

As shown in Table 2, the performance comparison for the left vs. right hand classification task reveals that the LSTM model outperforms both the Random Forest and CNN models in terms of accuracy, achieving 88.9%. The CNN model closely follows with an accuracy of 88%, while the Random Forest lags behind at 83%. When examining precision, the CNN leads with 77%, indicating a higher proportion of correctly identified positive results, whereas both the LSTM and Random Forest

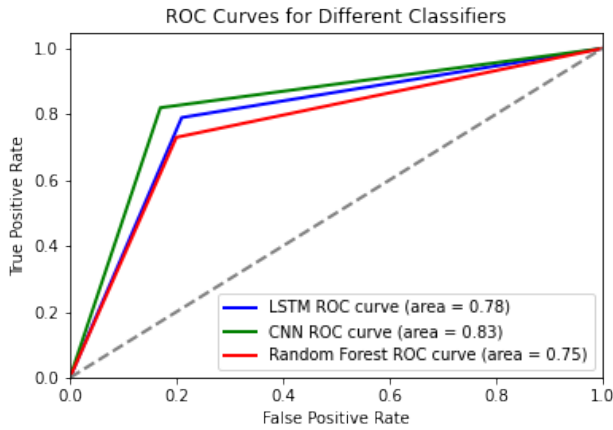


Fig. 5. ROC comparison of classifiers

TABLE II
COMPARISON OF PERFORMANCE METRICS FOR LEFT VS RIGHT

Performance Metrics	Random Forest	CNN	LSTM
Accuracy	83%	88%	88.9%
Precision	75%	77%	75%
Recall	74%	81%	80%
F1-score	74%	79%	77%

models achieve 75%. Recall is highest for the CNN at 81%, suggesting it is more effective at identifying true positives compared to the LSTM (80%) and Random Forest (74%). Finally, the F1-score, which balances precision and recall, is highest for the CNN at 79%, followed by the LSTM at 77%, and the Random Forest at 74%. These results indicate that while the LSTM offers the highest accuracy, the CNN provides a more balanced performance across precision, recall, and F1-score.

V. CONCLUSION

In our study, we compared the classification performance of machine learning and deep learning algorithms using EEG data gathered during a Braille stimulus task. The findings demonstrated a high level of accuracy in the binary classification of symbols related to the left and right hands, indicating the existence of distinct brain processes involved in Braille communication when using each hand. However, classification performance for intersymbol distinctions was only somewhat accurate, indicating a less pronounced difference in brain representations of specific Braille symbols. Across all categorisation tests, deep learning models consistently outperformed traditional machine learning methods, demonstrating their greater capacity to capture the complex neuronal dynamics associated with Braille reading.

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